

1.

Let the domain be the set of integers $\mathbb{Z}$.

What is the truth value of the statement

$$\forall x (x^2 \geq x)?$$

Solution:

The truth value of the statement
is True

2.

What is the logical negation of the statement

$$\forall x (P(x) \rightarrow Q(x))$$

Solution:

$$\neg [\forall x (P(x) \rightarrow Q(x))] \equiv \exists x (P(x) \wedge \neg Q(x))$$

3.

if $L(x, y)$ means "person x loves person y ," how do we translate the statement $\exists y \forall x L(x, y)$?

Solution:

There exists a person whom every one loves

4. Translate the following sentence into Formal Logic:

"No systems engineer lacks access permission."

(Domain: all employees, $E(x)$: systems engineer;
 $A(x)$: has access ~~the~~ permission).

Solution: $\neg \exists x (E(x) \wedge \neg A(x))$

or $\forall x (E(x) \rightarrow A(x))$

5. The logical negation of the statement

$\exists x \forall y P(x, y)$

Solution:

$\forall x \exists y \neg P(x, y)$

6. How to express: "Every file contains at least one security vulnerability"?

(Domain of x : files, domain of y : vulnerabilities
 $V(y, x)$: y exists in x).

Solution: $\forall x \exists y (V(y, x))$

7. Let the domain be the students $S = \{\text{David, Ruth}\}$.
if $P(\text{David})$ is true and $P(\text{Ruth})$ is false,
what is the truth value of.

$$\exists x \neg P(x)$$

Solution:

The truth value is True

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